

FINAL SCAN SHEET

Every ★ HIGH YIELD (14) and △ TRAP (8) from the guide — nothing else. Five minutes, closed book, right before you walk in.

★ HIGH YIELD — things he explicitly flagged

1. **Lymph is also connective tissue.** He said “blood and also lymph” — a matching question could pair either one.
2. The tube *is* the argument. RBCs occupy 45% of blood volume; every WBC and platelet in your body fits in the <1% buffy coat. That's why RBC count — and nothing else — sets viscosity.
3. **Plasma proteins are NOT used by cells as fuel.** He lingered here. RBCs and WBCs sit directly in a protein-rich fluid and never touch it — they use glucose, fatty acids, O₂. This is a ready-made true/false question.
4. **The RBC carries oxygen but never uses it.** He called this the elegant part of the design — the cell responsible for all the oxygen doesn't touch it. Lacking mitochondria is *why*. Expect this as a “which is true” item.
5. **“Once committed to a pathway, they cannot change.”** A myeloid progenitor can never become a lymphocyte. He said this almost verbatim — it's phrased like a true/false item.
6. **EPO comes from the KIDNEYS** (liver to a small extent), and **hypoxia** is the trigger. Kidney → EPO → red marrow → more RBCs → O₂ restored → stimulus off.
7. **WBCs are the only TRUE cells among the formed elements** — nucleus plus every organelle. RBCs have no nucleus; platelets aren't even whole cells. He stressed this phrasing, and “true cell” is precise exam language.
8. **Eosinophilia — he named this as licensing-exam material.** Eosinophils above the normal 2–4% points to **parasitic infection, allergies, or asthma**. Think eczema, allergic rhinitis, seasonal allergy — and note it may only be raised *during* a reaction, not permanently.
9. **Platelets are not cells.** They are **cytoplasmic fragments** broken off a much larger marrow cell, the **megakaryocyte** (mega = large). No nucleus → they age fast → removed within ~10 days if unused.
10. **Vascular spasm is “probably the most important factor”** — his words — because it is immediate and cuts flow to the injury before anything else can happen. But it is never sufficient alone; a small leak persists, which is why steps 2 and 3 exist.
11. **Press and hold. Never massage.** The platelet plug and fibrin mesh form within the first minute and are **still fragile**. Massage tears them → re-bleeding → and here's the detail he added: **you may see nothing at the time and a big bruise the next day**. Press ~30 seconds; once bleeding stops, you're done.
12. **Thrombus vs embolus is a definition pair, and the difference is motion.** Thrombus = forms and **persists in an unbroken vessel** — no trauma. Embolus = the same clot once it **travels**. DVT is the classic thrombus.
13. **Why does liver failure cause bleeding?** He asked this deliberately, and a student guessed erythropoietin (wrong — that's kidney). The chain: **clotting factors are proteins → the liver makes proteins → severe liver disease → no fibrinogen or prothrombin → bleeding**. This connects straight back to §2. Expect it.
14. **Newborns get a vitamin K injection at birth.** Vitamin K is made by **gut bacteria**, and a newborn's gut hasn't been colonised yet. It's prophylaxis — so that any injury doesn't become fatal.

⚠ TRAP — misconceptions he corrected in class

- 1. Thick blood does not cause atherosclerosis.** A student asked this directly; he said atherosclerosis is **plaque buildup** (lipid). Viscosity affects how easily blood *moves*; plaque is a separate mechanism. Don't let a question merge them.
- 2. A high reticulocyte count is not a disease.** A student asked; he was firm: it is a **compensatory mechanism**. The marrow is responding correctly to a problem elsewhere — bleeding, hemolysis, or iron finally arriving. The pathology is somewhere else.
- 3. Thyroid does not directly drive erythropoiesis.** Another student asked. Thyroid sets *cellular metabolic rate*; any link to RBC production is **indirect**.
- 4. Stress alone can raise your WBC.** He said that when a count sits at the high end, he asks about the day of the draw — a stressful day can naturally double it. A high WBC is not automatically infection.
- 5. WBC count alone CAN indicate sepsis.** A student asked whether you must wait for culture. His answer: no — “definitely alone can be indicator.” Sepsis kills in hours via blood pressure collapse, so clinicians start **broad-spectrum antibiotics immediately** and let culture refine it later.
- 6. NK cells DO have granules** — they're classed as agranulocytes only because the granules aren't visible under light microscopy. He raised this unprompted. Also note he corrected a student who thought all five types were granulocytes: **lymphocytes and monocytes are agranulocytes**.
- 7.** A student asked whether more nitric oxide is therefore better. His answer: **no — the normal amount**. “Nothing to say the higher is the better or the lower is the better. Just always the normal amount.” Beware any option phrased as “more X is better.”
- 8. Aspirin does not destroy platelets**, and no blood thinner touches RBCs. He was explicit: “not destroying platelets, just the ability for aggregation kind of reduces.” The count stays normal; the stickiness drops.

If any of these don't ring a bell instantly — go find it in the full guide tonight, not tomorrow at 6:30.